



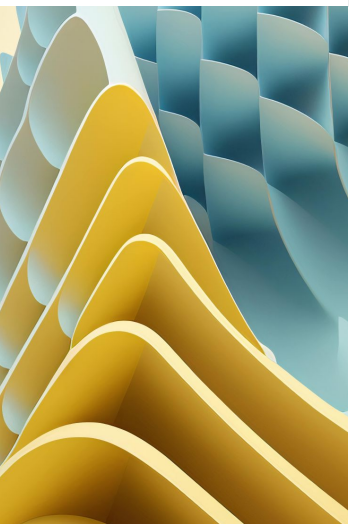
Grid Dynamics

trusted engineering partner for digital transformation

Principles of System Design and Solution Architecture of Intelligent Digital Platforms



Understanding a System and System Design



What is a System?

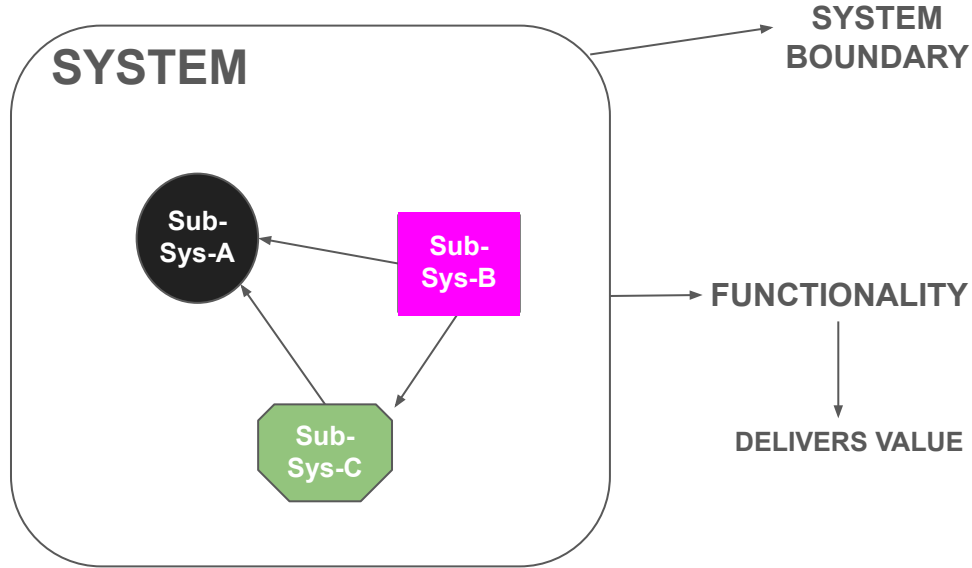
A system is a group of interacting or interrelated elements that act according to a set of rules to form a unified whole.

- A collection of interconnected components—hardware, software, or both—that work together toward a common goal, such as processing, storing, and managing information.
- These components include input devices, processors, memory, output devices, software, and networking elements, all functioning in harmony to receive input, perform computations, store data, and generate output.

Key characteristics of a system include:

- Structure: The organized arrangement of components.
- Function: The purpose or objective the system is designed to achieve.
- Interconnectivity: The way components interact and depend on each other.
- Boundary: The limits that define what is inside or outside the system.

System and its Construct?



KEY TENETS:

- SYSTEM EMERGENT PROPERTY
- SYSTEM- BENEFACTORS
- SYSTEM- VALUE
- VALUE- DEFINITION
- NEED-INTENT-PROCESS-FORM

What is System Design?

System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements.

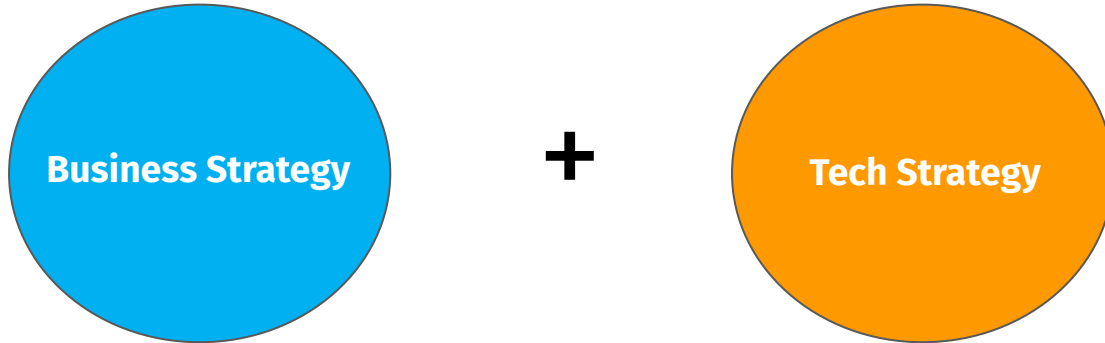
It involves organizing the system's elements and their relationships to ensure the system meets its intended goals efficiently and effectively.

System design typically covers:

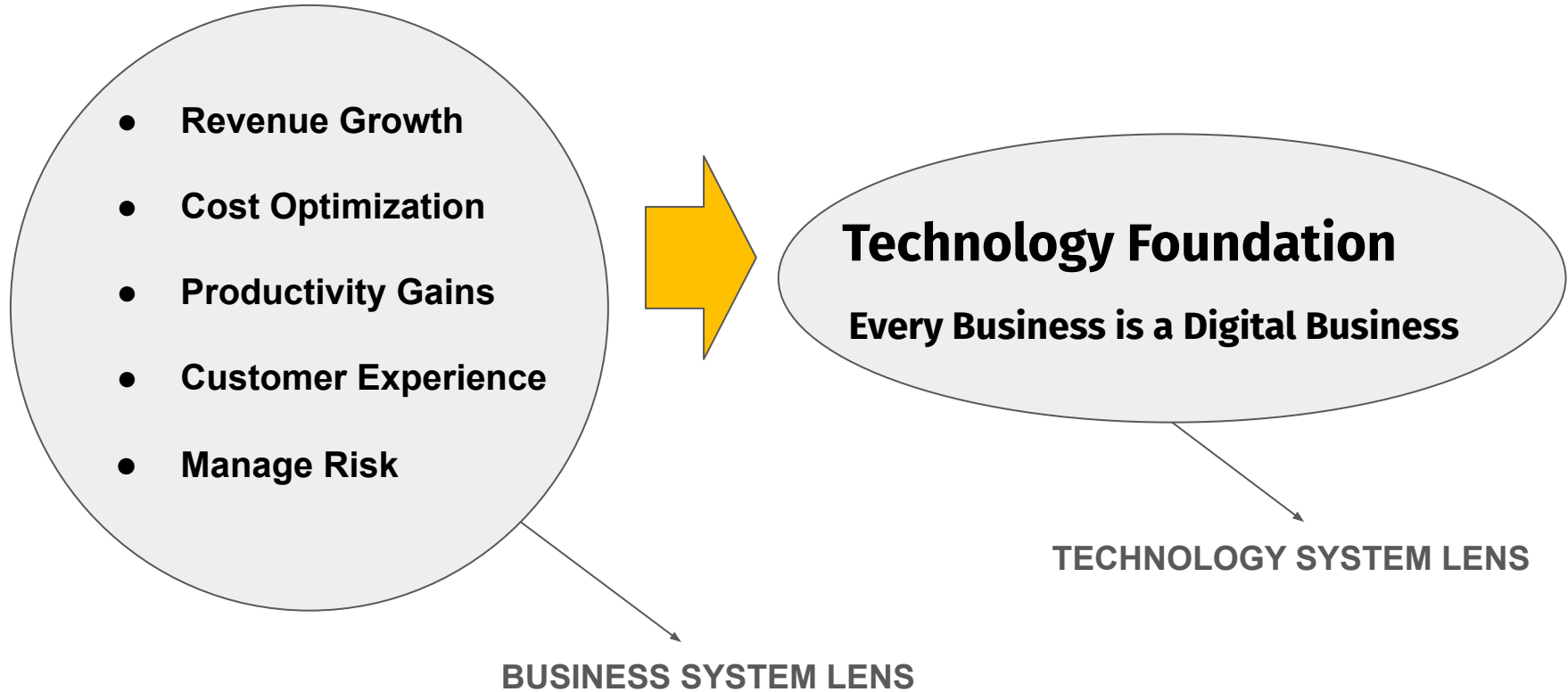
- Identifying system components and their interactions.
- Defining input, processing, control, output, and feedback mechanisms.
- Establishing how the system will operate within its environment and interact with other systems.
- Ensuring interoperability, scalability, reliability, and maintainability of the system (**System Illities**)

Enterprise Architecture

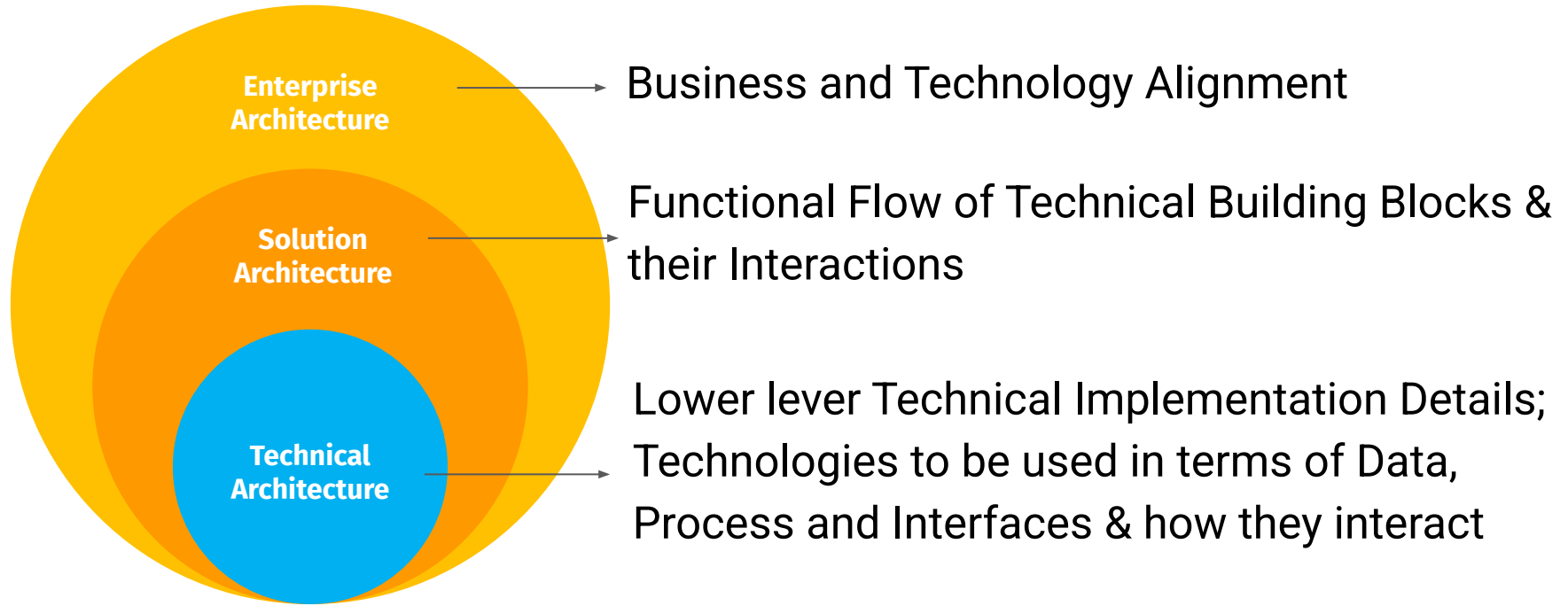
Enterprise architecture is the process by which organizations standardize and organize Technology building blocks to align with business goals.



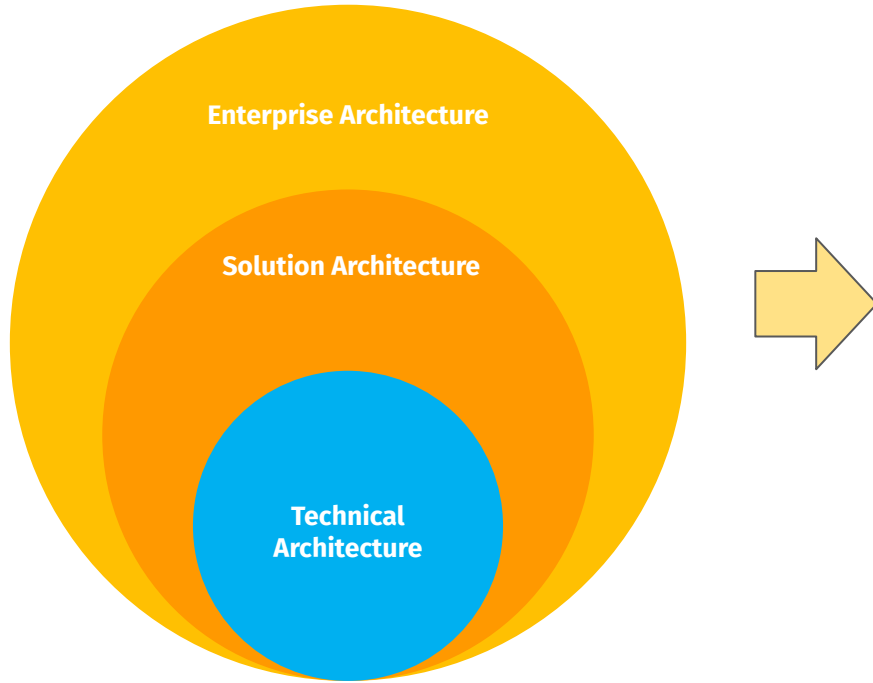
Business Imperatives: For whom the bell tolls



Architecture Abstraction-Varying Perspectives-Same Story



A Systems Thinking Approach

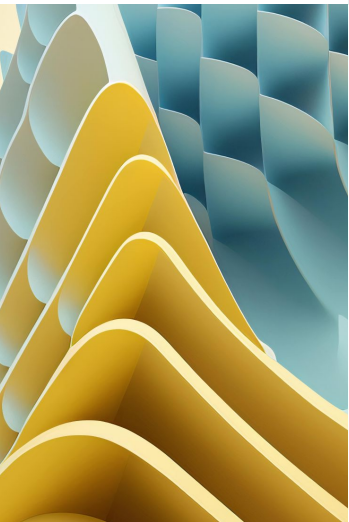


- ✓ **Interconnected Set Of Elements**
- ✓ **Organized Coherently to achieve a goal**
- ✓ **3 Components**
 - Elements
 - Interconnections
 - Function or Purpose

Emergent Property of a System



Architecting a Solution for Business Value



The State of Large Enterprises

- ★ Enterprises are a a entangled ecosystem- M&A's, Legacy systems wired together for immediate short term gains
- ★ Hampers business agility, change is hard, time consuming & very costly
- ★ Difficult to innovate and experiment and infuse new ideas to disrupt the market or maintain competitive advantage

The Digital & AI first companies are leaner and are **able to go to the market much faster**; most of them are also **adopting cloud-native paradigm**

Intelligent Digital Platforms & Cloud Native Applications

- ★ Digital Platforms that embrace intelligence (via AI/ML) either from scratch or as an after-thought
- ★ Cloud-native is a construct that uses the managed service abstraction of the public cloud providers
- ★ The end goal is to derive business value in terms of Revenue growth, Cost reduction, Productivity enhancement or improving Customer experience & loyalty and ensure business agility

The Focus on end consumers

As an enterprise platform - it has many consumers hooked **via API's** - so It's imperative **to engage with the end customers** and design their experiences with **AI-in-the-loop in today's large enterprises**

Explaining the terms that Matter!

Architecting/Architecture:

Convert upstream ambiguity to downstream clarity

Get to an implementation blueprint for the dev team to implement

Cloud Native:

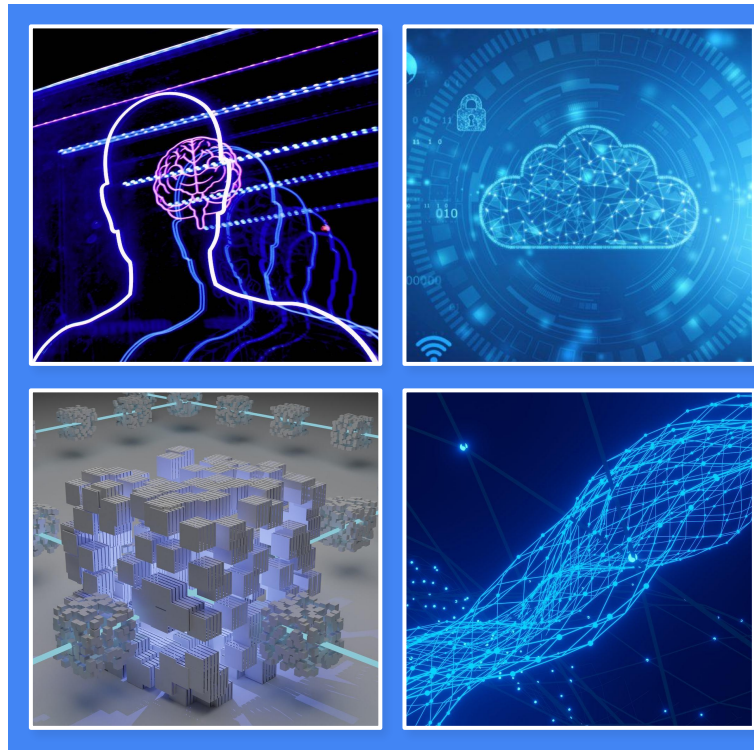
Be deployable across any cloud- CI/CD, Containers, Microservices- also story of scale, BCP/DR, Agility, etc.

Intelligent:

A trained machine learning or deep learning model as part of the solution or application architecture building block- They amplify business outcomes

Digital Platforms:

The components of the system that play a part in the business outcomes, can be accessed easily & makes future developments easier



Product Development Services

Forrester defines PDS 2.0 as a:

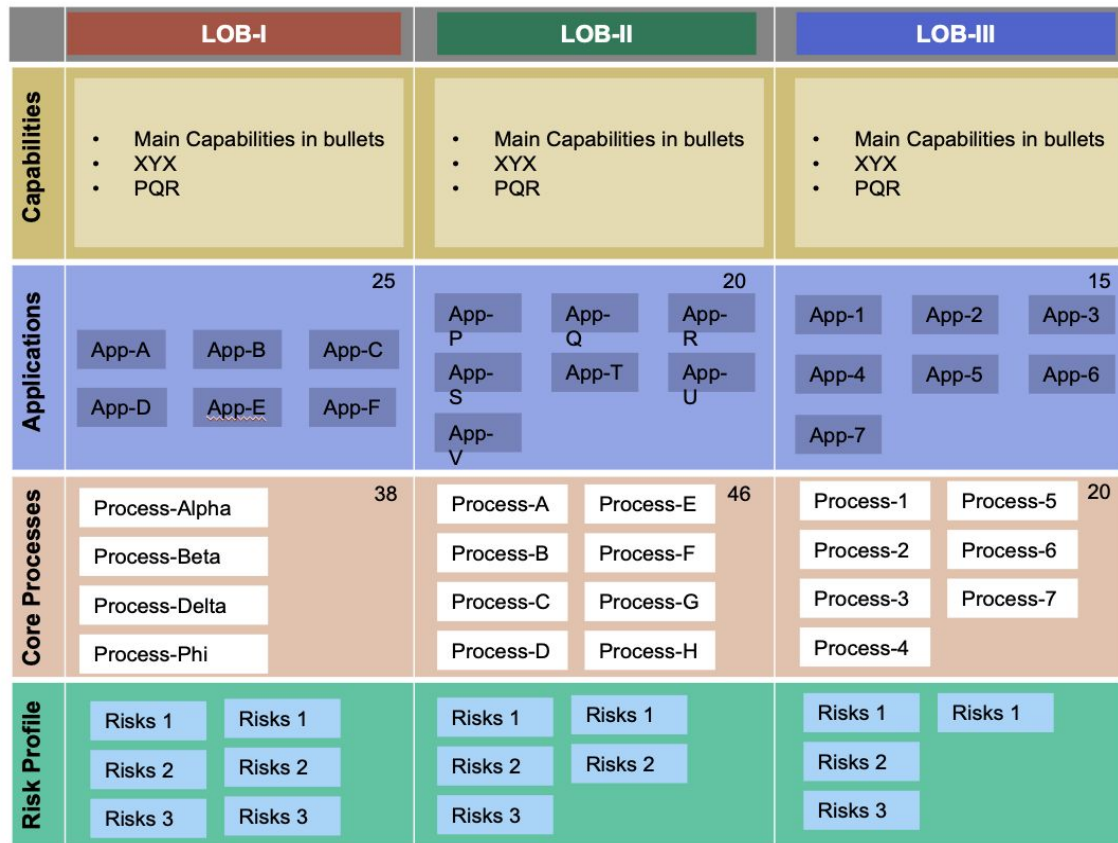
“Proactive, solution-based, domain-led offering based on a portfolio of internally and externally developed assets and aimed at improving the client's ability to innovate and cut time to market.”

Source: Forrester.com

Product Management Focus:

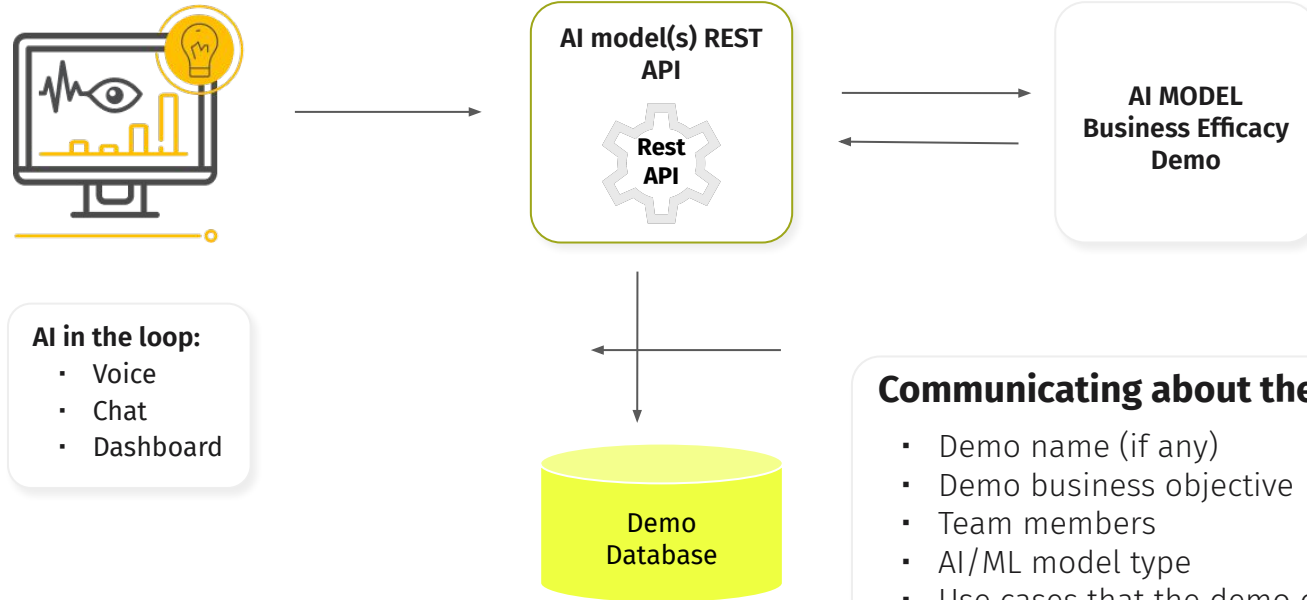
- ✓ Launching new capabilities and platforms
- ✓ Improve Time-To-Market
- ✓ Build and Maintain large Product Portfolios at reduced costs
- ✓ Manage Evolution of Product/Platform Functionality by adding technical debt in terms of features as long as it provides value to the end customers (internal or external as the case may be)

Enterprise Architecture Representation



- **Separation of Concerns**
- **Line of Business Mapped to Capabilities-Applications-Core Processes-Risks**
- **Always updated as Layers evolve**

Solution Architecture Representation

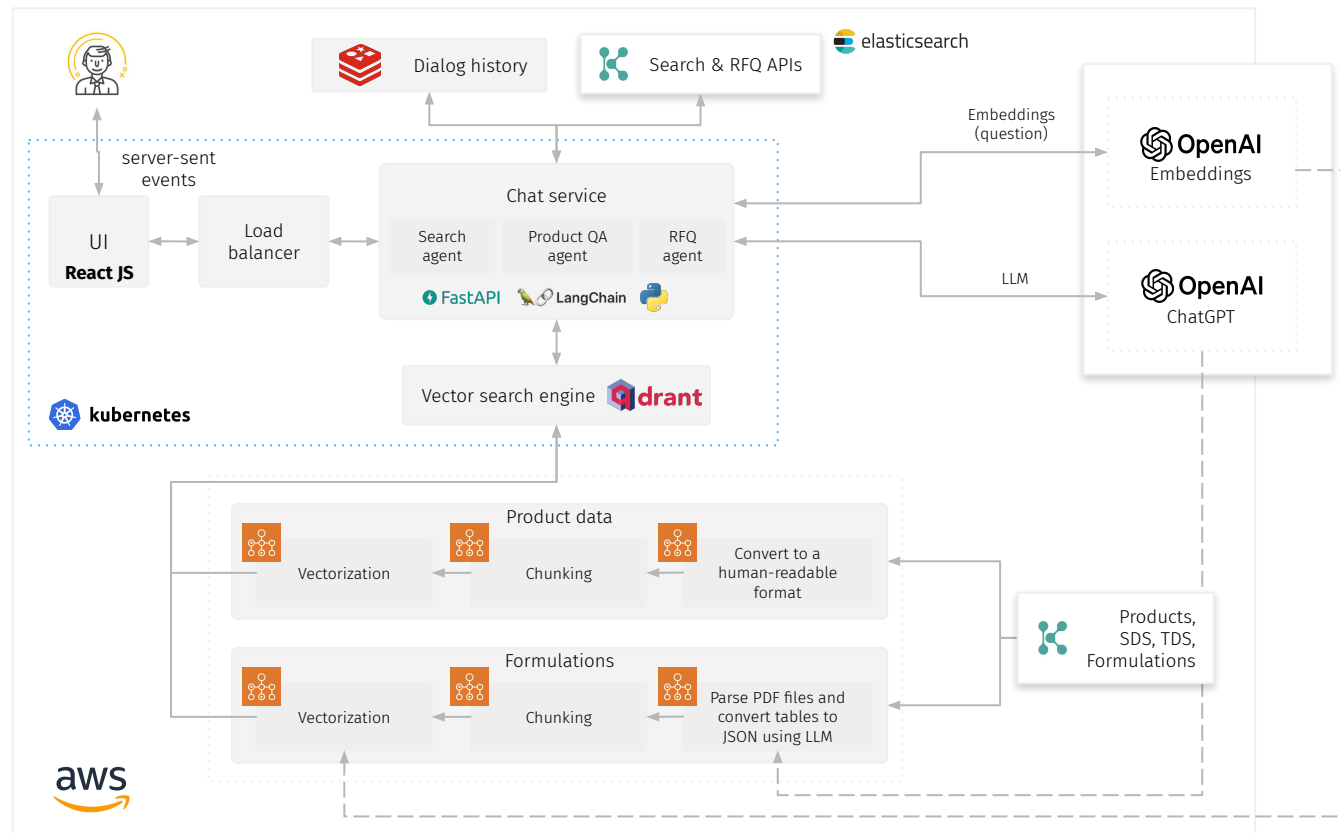


Communicating about the demo: A template

- Demo name (if any)
- Demo business objective
- Team members
- AI/ML model type
- Use cases that the demo can be applied to
- Model Metadata: Training time, Infra requirements
- Front and Back end technology details
- Other highlights

Technical Architecture Representation

- **System based on Chain-of-Thought and Retrieval Augmented Generation approaches**
- **Several AI agents that are responsible for key parts of the conversation:**
Search, Product QA, RFQ
- However this **GenAI based data extraction and harmonization** can however be very expensive



Foundational Building Blocks of any Technical Solution

Artificial Intelligence (AI)

Generative AI

Optimization

Search

ML
(Statistical Modeling)

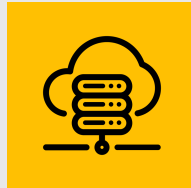
Deep Learning
(Neural Networks)

Reinforcement
Learning



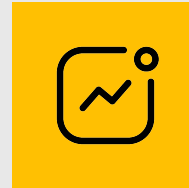
Cloud

Advanced Expertise in building Modern Cloud Platforms



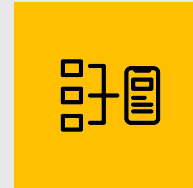
Data

Extensive hands-on experience in building Data Platforms



Analytics

Domain Specific workflows across GigaCube Industries

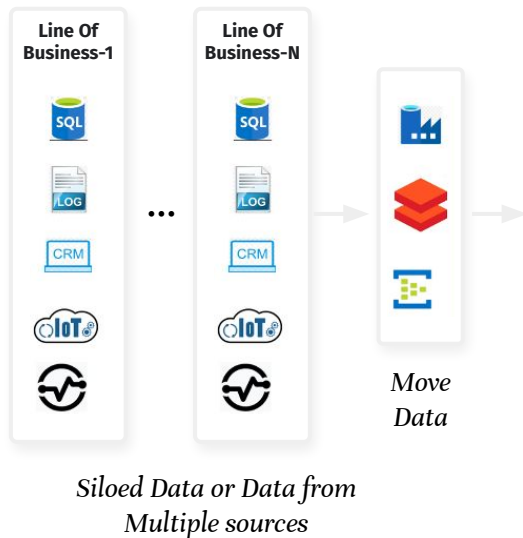


User Experience

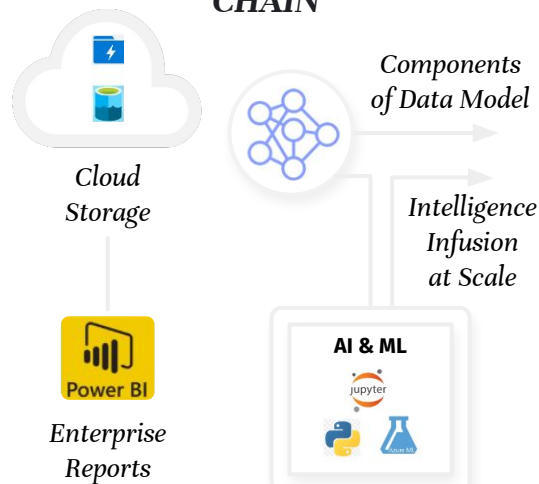
UX Design in the era of Human-Machine Interactions

Enterprise Digital Continuum- Symphony of many Platforms

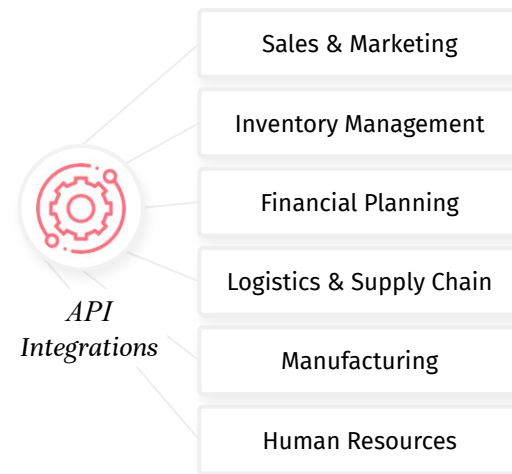
STRATEGIC INTENT



CLOUD FOR SCALE & INFUSING AI INTO THE BUSINESS VALUE CHAIN



DIGITAL ECONOMY POWERED BY API'S TO HARNESS THE POWER OF INTELLIGENCE



AI/Gen AI Infusion, Cybersecurity, Governance, Accessibility, Micro-Frontends and SRE/AIOps



Thank You !

